



# *Clostridioides difficile* infection surveillance in Belgium: trends and insights from 2013 to 2024

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## SUMMARY

**Background:** *Clostridioides difficile* remains the leading cause of infectious diarrhoea in healthcare settings and is an urgent public health threat. In Europe, the epidemiology of *C. difficile* infection (CDI) has evolved significantly over the past decades.

**Aim:** This study aimed to provide an updated overview of CDI epidemiology in Belgium between 2013 and 2024.

**Methods:** Surveillance data were collected from acute hospitals, reporting at least one semester annually and submitting five consecutive samples to the National Reference Centre. Cases occurring 48 h or more after admission were classified as hospital-associated *C. difficile* infection (HA-CDI). Microbiological analyses included ribotyping, toxin profiling, and antimicrobial susceptibility testing.

**Findings:** Over the 12-year period, hospital participation remained high, although the number of samples declined. Patient characteristics and recurrence rates remained stable, while the proportion of HA-CDI and complicated cases decreased. CDI incidence increased from 2.24 cases per 10,000 patient-days in 2013 to 2.77 cases in 2024. Strain diversity remained substantial, with over 60 ribotypes (RTs) identified annually. Fifteen RTs accounted for 70% of isolates, with RT014, 020, 002, and 078 alone representing one-third of all strains since 2013, while RT106 and RT154 have emerged more recently. A key finding was the apparent eradication of the hypervirulent RT027 strain. Some less frequent RTs were associated with severe outcomes. All isolates remained susceptible to CDI treatments available in Belgium.

**Conclusion:** This study highlights major trends in CDI epidemiology including changes in incidence, clinical severity, and RT distribution. Given the epidemiological burden of CDI, continued efforts to monitor CDI evolution are warranted.

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## Introduction

Forty years after being identified as the causative agent of pseudomembranous colitis, *Clostridioides difficile* remains the leading cause of infectious diarrhoea in healthcare settings and is classified as an urgent public health threat by the European Centre for Disease Control and Prevention (ECDC). *C. difficile* infections (CDIs) have significant clinical, social, and economic consequences due to increased duration of hospitalisation, re-admission, treatment, laboratory diagnosis, and infection control measures [1,2]. Across Europe, CDIs account for 5.9% of all healthcare-associated infections [3], leading to over €3 billion in healthcare costs [4,5], and approximately 8382 attributable deaths [6,7].

CDI pathogenesis is primarily driven by enterotoxin A (TcdA) and cytotoxin B (TcdB), which are encoded by the *tcdA* and *tcdB* genes, respectively. Some *C. difficile* isolates also produce a binary toxin, encoded by *cdtA* and *cdtB* genes [5]. CDI's clinical spectrum ranges from mild diarrhoea to life-threatening toxic megacolon. Risk factors include advanced age, prolonged hospitalisation, and antimicrobial exposure, which disrupts gut microbiota and facilitates colonisation and infection by *C. difficile* [1]. The European Society of Clinical Microbiology and Infectious Diseases (ESCMID) publishes CDI treatment guidelines [5,8].

Since 2000, shifts in CDI's clinical presentation have been reported, including an increase in severe cases, community-acquired (CA) infections among individuals without traditional risk factors, and large-scale outbreaks in North America and Europe. These changes were largely driven by the emergence of hypervirulent strains, particularly ribotype (RT) 027 [1], whose global spread was facilitated by fluoroquinolone resistance [9]. Following the recognition of European outbreaks linked to RT027 emergence in 2005, the ECDC encouraged national CDI surveillance [5,10] and published a protocol in May 2015 [11].

In Belgium, RT027 was first identified during a hospital outbreak in 2005 [5], prompting the establishment of a national CDI surveillance in acute care hospitals. Annual participation in this surveillance became mandatory for all general hospitals in Belgium in 2007 and remained so until 2014, after which it became optional surveillance [12]. This surveillance system relies on epidemiological data submitted to the Scientific Institute of Public Health (Sciensano) and microbiological samples sent to the National Reference Centre (NRC) for *C. difficile*. For this article, we relied on data from the national CDI surveillance to provide an updated overview of CDI epidemiology in Belgium from 2013 to 2024.

## Methods

### Hospital-based surveillance system

Participating hospitals are required to report all identified CDIs and send five consecutive samples to the NRC for at least one semester per year. From 2015 onwards, participation became voluntary. General hospitals (primary), those with a university character (secondary), and those affiliated with a university (tertiary) participate in the surveillance. Epidemiological data (sex, age, presumed case origin, recurrent case, complicated case, patient status, and speciality ward) have been registered in the secured Healthdata (HD) platform since mid-2017. Patient

data can be linked to the samples with an automatically generated code. Denominator data (patient-days, admissions) are collected in a separate registry. Since 2017, our protocol has been aligned with the enhanced ECDC surveillance protocol [11].

### Case definitions

A CDI case is defined as a patient meeting one or more of the following criteria [12]: (1) diarrhoea or toxic megacolon and a positive laboratory test for *C. difficile* toxin A and/or B in the stools or a toxin-producing strain identified by culture or another method; (2) pseudomembranous colitis identified by lower gastrointestinal tract endoscopy; (3) histopathological findings characteristic of *C. difficile* in the colon (with or without diarrhoea) on a biopsy obtained during endoscopy, colectomy, or autopsy. Hospital-associated (HA) CDIs occur in patients presenting symptoms 48 h or more after hospital admission. All other CDIs that were not HA-CDI were referred to as non-HA-CDI in this paper. Presumed origin of CDI can be a healthcare setting (onset 48 h or more after admission, or up to 4 weeks post discharge), other healthcare setting (e.g. outpatient care), the community (symptoms start outside of a healthcare setting or within 48 h of admission, with no stay in the previous 12 weeks), undetermined (symptoms onset between 4–12 weeks after the last stay), or unknown. Recurrent CDI refers to a second infection occurring 2–8 weeks after symptom onset of the previously resolved infection. Complicated CDI cases are defined as patients (1) admitted to a healthcare facility for treatment of CA-CDI; (2) admitted to intensive care for CDI treatment; (3) who underwent a surgical procedure for toxic megacolon, perforation, or refractory colitis; or (4) who died due to CDI within 30 days of diagnosis.

### Microbiological surveillance

According to the surveillance protocol, hospitals should send a maximum of five consecutive samples per semester, but some submit more. In this study, we included all samples received. Isolates or faecal samples were processed for *C. difficile* culture, polymerase chain reaction (PCR) ribotyping, and determination of toxin status by cell culture cytotoxicity assay. Analyses were performed following ISO 15189 standards. PCR ribotyping was performed by amplifying 16S–23S rRNA intergenic regions, with amplicons analysed by high-resolution capillary electrophoresis [13]. RTs were assigned using the EU reference collection to ensure cross-study comparability. Each distinct RT and, since November 2022, all isolates were also screened by multiplex PCR for the presence of the *tcdA* and *tcdB* genes, the binary toxin genes (*cdtA* and *cdtB*), as well as the regulatory gene *tcdC*, including screening for specific deletions such as a 1-bp deletion at position 117 and larger deletions of 18 bp, 39 bp, and 54 bp [14,15]. In this study, 'hypervirulent' strains refer to RTs carrying *tcdA* and *tcdB* together with binary toxin genes and *tcdC* deletions.

*In vitro* susceptibility to metronidazole, vancomycin, and fidaxomicin was determined by measuring minimum inhibitory concentrations (MICs) by an agar dilution method following the European Committee on Antimicrobial Susceptibility Testing guidelines (v14.0). For susceptibility to moxifloxacin, clindamycin, erythromycin, rifampicin, tetracycline, and chloramphenicol, the disk diffusion method on Brucella agar was used, following the Comité de l'Antibiogramme de la Société Française de Microbiologie guidelines (V1.0 June 2024).

Table I

*Clostridioides difficile* infections (CDI) characteristics reported by hospitals that provided numerators for at least one semester, Belgium, 2013–2024

| CDI characteristic (%)                           | 2013 <sup>a</sup> | 2014 <sup>a</sup> | 2015 <sup>a</sup> | 2016 <sup>a</sup> | 2017        | 2018        | 2019        | 2020        | 2021        | 2022        | 2023        | 2024        |
|--------------------------------------------------|-------------------|-------------------|-------------------|-------------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| <b>Sex</b>                                       |                   |                   |                   |                   |             |             |             |             |             |             |             |             |
| Male                                             | 1159 (43.6)       | 1102 (45.7)       | 1240 (41.8)       | 1193 (42.6)       | 1130 (43.7) | 1053 (43.1) | 1025 (44.6) | 849 (42.5)  | 831 (44.9)  | 961 (44.5)  | 1051 (45.9) | 1072 (46.0) |
| Female                                           | 1500 (56.4)       | 1308 (54.3)       | 1730 (58.2)       | 1606 (57.3)       | 1453 (53.2) | 1383 (56.6) | 1267 (55.2) | 1149 (57.5) | 1021 (55.1) | 1195 (55.3) | 1236 (53.9) | 1259 (54.0) |
| Unknown                                          | 0 (0)             | 0 (0)             | 0 (0)             | 3 (0.1)           | 3 (0.1)     | 6 (0.2)     | 5 (0.2)     | 1 (0.1)     | 1 (0.1)     | 5 (0.2)     | 5 (0.2)     | 1 (0.04)    |
| <b>Age groups</b>                                |                   |                   |                   |                   |             |             |             |             |             |             |             |             |
| 0–64 years                                       | 704 (26.5)        | 681 (28.3)        | 795 (26.8)        | 736 (26.3)        | 725 (28.0)  | 689 (28.2)  | 661 (28.8)  | 527 (26.4)  | 564 (30.4)  | 567 (26.2)  | 669 (29.2)  | 694 (29.8)  |
| 65–79 years                                      | 747 (28.1)        | 710 (29.5)        | 807 (27.2)        | 776 (27.7)        | 724 (28.0)  | 754 (30.9)  | 715 (31.1)  | 628 (31.4)  | 604 (32.6)  | 753 (34.8)  | 790 (34.5)  | 838 (35.9)  |
| >80 years                                        | 1208 (45.4)       | 1019 (42.3)       | 1368 (46.1)       | 1290 (46.0)       | 1137 (44.0) | 999 (40.9)  | 920 (40.1)  | 844 (42.2)  | 685 (37.0)  | 841 (38.9)  | 833 (36.3)  | 800 (34.3)  |
| <b>HA-CDI<sup>b</sup></b>                        | 1584 (59.6)       | 1434 (59.5)       | 1771 (59.6)       | 1623 (57.9)       | 1509 (56.1) | 1488 (55.5) | 1339 (53.4) | 1200 (56.2) | 1124 (54.6) | 1376 (58.1) | 1407 (55.9) | 1396 (53.8) |
| <b>Presumed origin of HA-CDI<sup>b</sup></b>     |                   |                   |                   |                   |             |             |             |             |             |             |             |             |
| Community                                        | NA                | NA                | NA                | NA                | 18 (1.2)    | 33 (2.2)    | 43 (3.2)    | 45 (3.8)    | 37 (3.3)    | 48 (3.5)    | 73 (5.2)    | 63 (4.5)    |
| Acute care facility                              | 1584 (100.0)      | 1434 (100.0)      | 1771 (100.0)      | 1623 (100.0)      | 1408 (93.3) | 1309 (88.0) | 1166 (87.1) | 1040 (86.7) | 969 (86.2)  | 1153 (83.8) | 1094 (77.8) | 1047 (75.0) |
| Long-term care facility                          | NA                | NA                | NA                | NA                | 27 (1.8)    | 49 (3.3)    | 50 (3.7)    | 40 (3.3)    | 48 (4.3)    | 60 (4.4)    | 37 (2.6)    | 29 (2.1)    |
| Other healthcare facility                        | NA                | NA                | NA                | NA                | 7 (0.5)     | 3 (0.2)     | 4 (0.3)     | 3 (0.3)     | 4 (0.4)     | 6 (0.4)     | 10 (0.7)    | 7 (0.5)     |
| Undetermined                                     | NA                | NA                | NA                | NA                | 10 (0.7)    | 27 (1.8)    | 23 (1.7)    | 23 (1.9)    | 30 (2.7)    | 37 (2.7)    | 30 (2.1)    | 19 (1.4)    |
| Unknown                                          | NA                | NA                | NA                | NA                | 39 (2.6)    | 67 (4.5)    | 53 (4.0)    | 49 (4.1)    | 36 (5.2)    | 72 (5.2)    | 163 (11.6)  | 231 (16.5)  |
| <b>Presumed origin of non-HA-CDI<sup>b</sup></b> |                   |                   |                   |                   |             |             |             |             |             |             |             |             |
| Community                                        | 664 (61.8)        | 616 (63.1)        | 809 (67.5)        | 731 (62.0)        | 723 (61.3)  | 705 (59.1)  | 653 (55.8)  | 499 (53.4)  | 512 (54.8)  | 634 (64.0)  | 642 (57.7)  | 679 (58.0)  |
| Acute care facility                              | 164 (15.3)        | 167 (17.1)        | 183 (15.3)        | 189 (16.0)        | 233 (19.8)  | 233 (19.5)  | 205 (17.5)  | 159 (17.0)  | 185 (19.8)  | 158 (16.0)  | 173 (15.6)  | 179 (15.3)  |
| Long-term care facility                          | 113 (10.5)        | 82 (8.4)          | 104 (8.7)         | 124 (10.5)        | 77 (6.5)    | 75 (6.3)    | 71 (6.1)    | 62 (6.6)    | 53 (5.7)    | 53 (5.4)    | 60 (5.4)    | 62 (5.3)    |
| Other healthcare facility                        | NA                | NA                | NA                | NA                | 28 (2.4)    | 29 (2.4)    | 27 (2.3)    | 22 (2.4)    | 26 (2.8)    | 28 (2.8)    | 20 (1.8)    | 47 (4.0)    |
| Undetermined                                     | 62 (5.8)          | 53 (5.4)          | 38 (3.2)          | 59 (5.0)          | 40 (3.4)    | 66 (5.5)    | 54 (4.6)    | 40 (4.3)    | 43 (4.6)    | 40 (4.0)    | 59 (5.3)    | 52 (4.4)    |
| Unknown                                          | 72 (6.7)          | 58 (5.9)          | 65 (5.4)          | 76 (6.4)          | 78 (6.6)    | 84 (7.0)    | 160 (13.7)  | 152 (16.3)  | 116 (12.4)  | 77 (7.8)    | 158 (14.2)  | 151 (12.9)  |
| <b>Recurrent CDI</b>                             |                   |                   |                   |                   |             |             |             |             |             |             |             |             |
| Yes                                              | 251 (9.4)         | 209 (8.7)         | 313 (10.5)        | 311 (11.1)        | 272 (10.1)  | 281 (10.5)  | 259 (10.3)  | 189 (8.9)   | 207 (10.1)  | 253 (10.7)  | 264 (10.5)  | 273 (10.6)  |
| No                                               | 1942 (73.0)       | 1781 (73.9)       | 2166 (72.9)       | 2050 (73.2)       | 2123 (79.0) | 2162 (80.7) | 2003 (79.8) | 1778 (83.3) | 1642 (79.8) | 1814 (76.7) | 1841 (73.1) | 1840 (73.3) |
| Unknown                                          | 466 (17.5)        | 420 (14.4)        | 491 (16.5)        | 441 (15.7)        | 293 (10.9)  | 237 (8.8)   | 247 (9.8)   | 167 (7.8)   | 210 (10.2)  | 299 (12.6)  | 415 (16.5)  | 415 (16.2)  |
| <b>Complicated CDI</b>                           |                   |                   |                   |                   |             |             |             |             |             |             |             |             |
| Yes                                              | 576 (21.7)        | 532 (22.1)        | 725 (24.4)        | 644 (23.0)        | 415 (15.4)  | 172 (6.4)   | 172 (6.9)   | 153 (7.2)   | 150 (7.3)   | 209 (8.8)   | 245 (9.7)   | 243 (9.5)   |
| No                                               | 1529 (57.5)       | 1340 (55.6)       | 1612 (54.3)       | 1635 (58.4)       | 1841 (68.5) | 2210 (82.5) | 1913 (76.3) | 1672 (78.4) | 1683 (81.7) | 1831 (77.4) | 1848 (73.3) | 1899 (74.0) |
| Unknown                                          | 554 (20.8)        | 538 (22.3)        | 633 (21.3)        | 523 (18.7)        | 432 (16.1)  | 298 (11.1)  | 424 (16.9)  | 309 (14.5)  | 226 (11.0)  | 326 (13.8)  | 427 (17.0)  | 424 (16.5)  |
| <b>Patient status after CDI</b>                  |                   |                   |                   |                   |             |             |             |             |             |             |             |             |
| Alive                                            | 2039 (76.7)       | 1764 (73.2)       | 2259 (76.1)       | 2118 (75.6)       | 2028 (78.4) | 1963 (80.4) | 1866 (81.3) | 1646 (82.8) | 1449 (80.9) | 1712 (79.2) | 1763 (76.9) | 1833 (78.6) |
| Dead, related to CDI                             | 91 (3.4)          | 74 (3.1)          | 85 (2.9)          | 66 (2.4)          | 61 (2.4)    | 61 (2.5)    | 44 (1.9)    | 46 (2.3)    | 39 (2.1)    | 52 (2.4)    | 60 (2.6)    | 59 (2.5)    |
| Dead, unrelated to CDI                           | 337 (12.7)        | 321 (13.3)        | 394 (13.3)        | 375 (13.4)        | 349 (13.5)  | 338 (13.8)  | 353 (15.4)  | 277 (13.9)  | 241 (13.0)  | 294 (13.6)  | 256 (11.2)  | 257 (11.1)  |
| Unknown                                          | 192 (7.2)         | 251 (10.4)        | 232 (7.8)         | 243 (8.7)         | 148 (5.7)   | 80 (3.28)   | 33 (1.4)    | 30 (1.5)    | 74 (4.0)    | 103 (4.8)   | 213 (9.3)   | 183 (7.9)   |

| Speciality ward             |            |            |            |            |            |            |            |            |            |            |            |            |
|-----------------------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|
| Geriatrics                  | 816 (30.7) | 650 (27.0) | 963 (32.4) | 835 (29.8) | 763 (28.4) | 777 (29.0) | 662 (26.4) | 613 (28.7) | 575 (27.9) | 641 (27.1) | 637 (25.3) | 648 (25.3) |
| Other                       | 139 (5.2)  | 129 (5.4)  | 148 (5.0)  | 168 (6.0)  | 152 (5.7)  | 271 (8.1)  | 214 (8.5)  | 169 (7.9)  | 208 (10.1) | 249 (10.5) | 321 (12.7) | 273 (10.6) |
| Other medical department    | 357 (13.4) | 303 (12.6) | 316 (10.6) | 341 (12.2) | 302 (11.2) | 358 (13.4) | 322 (12.8) | 254 (11.9) | 239 (11.6) | 297 (12.6) | 264 (12.5) | 285 (11.1) |
| Gastroenterology            | NA         | NA         | 10 (0.3)   | 1 (0.0)    | 124 (4.6)  | 307 (11.5) | 259 (10.3) | 213 (10.0) | 174 (8.5)  | 267 (11.3) | 260 (10.3) | 262 (10.2) |
| Haematooncology             | 199 (7.5)  | 201 (8.3)  | 215 (7.2)  | 220 (7.9)  | 241 (9.0)  | 220 (8.2)  | 237 (9.5)  | 190 (8.9)  | 196 (9.5)  | 198 (8.4)  | 251 (10.0) | 319 (12.4) |
| Intensive care unit         | 153 (5.8)  | 141 (5.9)  | 179 (6.0)  | 169 (6.0)  | 165 (6.1)  | 180 (6.7)  | 176 (7.0)  | 134 (6.3)  | 164 (8.0)  | 155 (6.6)  | 190 (7.5)  | 187 (7.3)  |
| Surgery                     | 204 (7.7)  | 191 (7.9)  | 207 (7.0)  | 198 (7.1)  | 149 (5.5)  | 165 (6.2)  | 174 (6.9)  | 140 (6.6)  | 130 (6.3)  | 172 (7.3)  | 157 (6.2)  | 178 (6.9)  |
| Pneumology                  | 128 (4.8)  | 110 (4.6)  | 124 (4.2)  | 113 (4.0)  | 102 (3.8)  | 101 (3.8)  | 104 (4.2)  | 105 (4.9)  | 79 (3.8)   | 105 (4.4)  | 108 (4.3)  | 128 (5.0)  |
| Paediatrics                 | 69 (2.6)   | 79 (3.3)   | 85 (2.9)   | 80 (2.9)   | 77 (2.9)   | 77 (2.9)   | 75 (3.0)   | 29 (1.4)   | 77 (3.7)   | 66 (2.8)   | 99 (3.9)   | 123 (4.8)  |
| Cardiology                  | 76 (2.9)   | 78 (3.2)   | 74 (2.5)   | 69 (2.5)   | 85 (3.2)   | 69 (2.6)   | 79 (3.2)   | 64 (3.0)   | 58 (2.8)   | 76 (3.2)   | 76 (3.9)   | 57 (2.2)   |
| Nephrology                  | 90 (3.4)   | 107 (4.4)  | 131 (4.4)  | 88 (3.1)   | 119 (4.4)  | 86 (3.2)   | 66 (2.6)   | 62 (2.9)   | 61 (3.0)   | 64 (2.7)   | 72 (2.9)   | 74 (2.9)   |
| Obstetrics/gynaecology      | 11 (0.4)   | 10 (0.4)   | 11 (0.4)   | 9 (0.3)    | 5 (0.2)    | 7 (0.3)    | 5 (0.2)    | 3 (0.1)    | 2 (0.1)    | 5 (0.2)    | 5 (0.2)    | 10 (0.4)   |
| COVID-19 general department | NA         | NA         | NA         | NA         | NA         | NA         | NA         | 21 (1.0)   | 17 (0.8)   | 4 (0.2)    | 0 (0)      | 0 (0)      |
| COVID-19 ICU                | NA         | NA         | NA         | NA         | NA         | NA         | NA         | 1 (0.1)    | 8 (0.4)    | 0 (0)      | 0 (0)      | 0 (0)      |
| Missing                     | 417 (15.7) | 411 (17.1) | 507 (17.1) | 511 (15.2) | 404 (15.0) | 116 (4.3)  | 136 (5.4)  | 136 (6.4)  | 71 (3.4)   | 67 (2.8)   | 80 (3.2)   | 22 (0.9)   |

HA-CDI: hospital-associated *Clostridioides difficile* infection; ICU: intensive care unit; NA: not applicable.

Note: 'Dead, related to CDI' includes possible and confirmed causes of death.

<sup>a</sup> From 2013 to 2016, no distinction could be made between patients for the same admission. Starting from 2017, the Healthdata environment enabled us to deduplicate by admission and patient for the age and sex characteristics.

<sup>b</sup> HA-CDIs are calculated using start symptoms  $\geq 48$  h after admission. All other CDIs are defined as non-HA-CDIs. The variable 'presumed origin' is reported by the hospital.

## Data analysis

Non-recurrent CDI episodes were deduplicated per hospitalisation. Age groups, sex, and CDI deaths were deduplicated at the patient level. One-day hospitalisations were excluded from the datasets. Data validation, cleaning, and analysis were completed in HD using SAS Enterprise Guide 7.13. PCR RT profiles were analysed using Bionumerics v8.1.1 (Applied Maths, Belgium). Graphs were created with GraphPad Prism 9 (GraphPad Software Inc.), and statistical analysis was performed with R Version 4.3.3 (R Foundation for Statistical Computing). To assess the statistical significance of proportions, a z-test was used, while for the incidence rates, a Poisson rate test was applied. The 95% confidence intervals (CIs) were reported where appropriate.

## Results

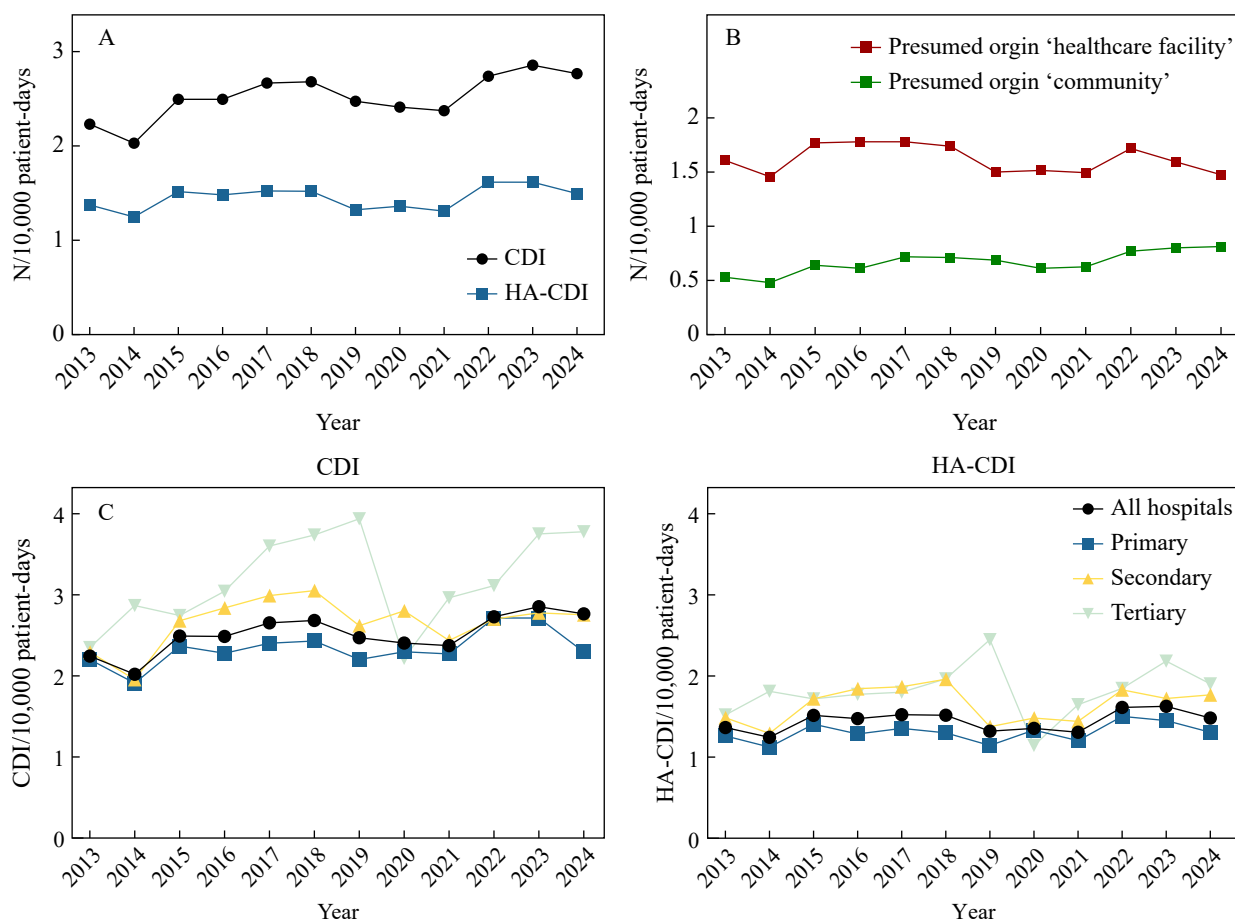
### Hospital participation and retrieved samples

After the surveillance became non-mandatory in 2015, participation decreased, though never below 80%. In 2024, participation reached an all-time high, with 91% of Belgian hospitals providing numerators and denominators for at least one semester. Although sending up to five strains to the NRC

per participating semester was mandatory, compliance was not always achieved. In 2024, only 59% of hospitals participating in the surveillance sent samples to the NRC ([Supplementary Figure A1](#)). The number of strains sent decreased in 2020 and remained low (1030 strains in 2019 vs 821 in 2024). Overall, 11,129 samples were collected between 2013 and 2024. Samples that could not be cultured and those that were non-toxicogenic were excluded from further analysis.

### Patient and case characteristics

Most patients with CDI were elderly women ([Table I](#)). To investigate the origin of the infections, the HA status of CDI cases (based on the standardised cut-off of 48 h or more after admission) was compared with the variable 'presumed case origin', encoded by the hospital. According to both sources, more than 50% of cases occurred in the hospital. However, the percentage of HA-CDI decreased over time. The overlap between the presumed origin 'acute care facility' and the calculated HA-CDI was 93.3% in 2017, with a decrease to 75.0% in 2024. The presumed origin of non-HA-CDIs data indicated where these infections were contracted, if not in the hospital. Most of the non-HA-CDIs originated from the community (58.0%), while the proportion of non-HA cases originating from a long-term care facility declined (10.5% in 2013 vs 5.3% in



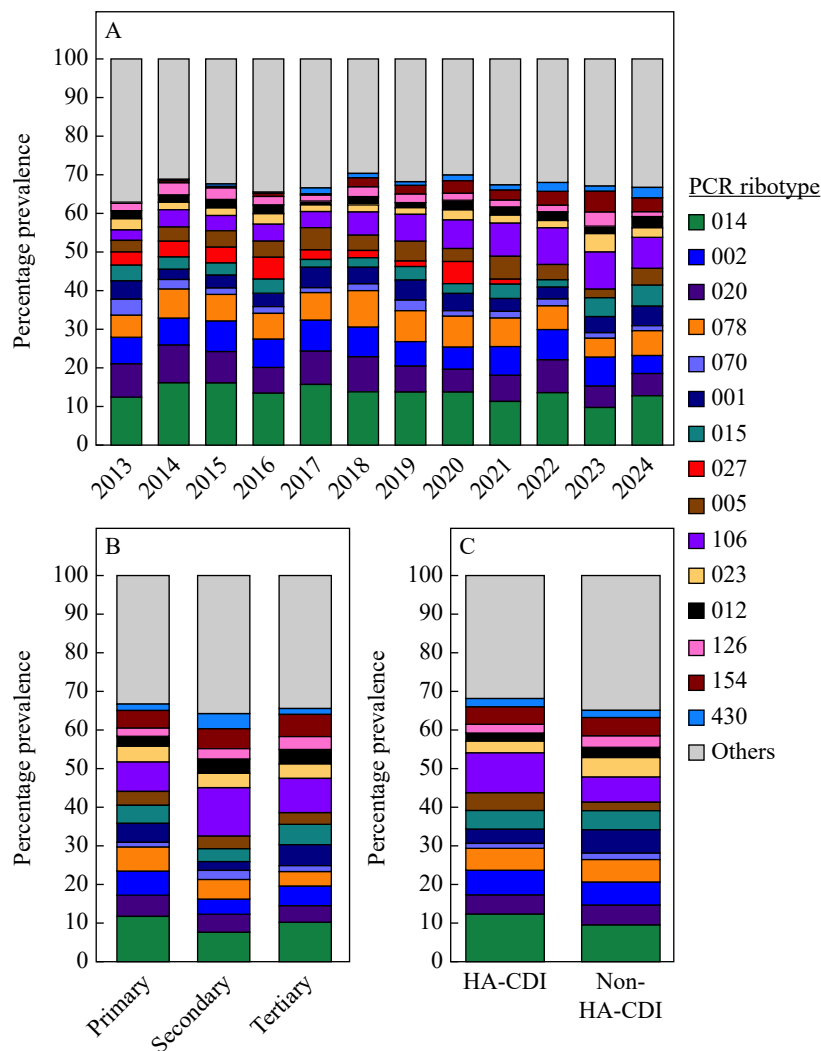
**Figure 1.** *Clostridioides difficile* infections (CDIs) and hospital-associated *Clostridioides difficile* infections (HA-CDIs) incidence (per 10,000 patient-days) (A), by type of hospital (C), and for presumed origin 'healthcare facility' and 'community' (B). Belgium, 2013–2024. Note: Only hospitals providing numerators and denominators for at least one semester were included. HA-CDI: onset of CDI 48 h or more after admission (calculated using infection date and admission date); presumed origin is reported by hospitals. 'Healthcare facility' includes cases originated from 'acute care facility', 'long-term care facility', and 'other healthcare facility'.

2024,  $z = 4.6$ ,  $P < 0.05$ ). The recurrence percentage fluctuated between 8.7% and 11.1% over the study period. Complicated infections decreased from 21.7% in 2013 to 9.5% in 2024 ( $z = 12.1$ ,  $P < 0.05$ ). The number of patients who died because of CDI was low (2.5% in 2024) (Supplementary Figure A2). Most CDI patients came from geriatrics wards, although this percentage decreased to 25.3% in 2024 compared with 30.7% in 2013 ( $z = 4.4$ ,  $P < 0.05$ ). There was an increasing number of unknown values for presumed origin, recurrence, complicated course, and patient status.

### CDI incidences

Since 2013, an upward trend has been observed in CDI incidence, with a significant increase post pandemic from 2.37 to 2.85 cases per 10,000 patient-days between 2021 and 2023. The same trends can be observed in the HA-CDI incidence;

however, it remained stable between 2022 and 2023 (1.62 cases per 10,000 patient-days). In 2024, CDI and HA-CDI incidence decreased slightly (Figure 1A, Supplementary Table A3). The incidence in tertiary hospitals was often higher than in primary and secondary hospitals. However, a steep and statistically significant decline in incidence was reported in 2020 in tertiary hospitals, with the incidence of CDI and HA-CDI per 10,000 patient-days dropping from 3.93 and 2.46 in 2019 to 2.23 and 1.16 in 2020, respectively. After 2020, the incidence in tertiary hospitals increased again and remained higher than in primary and secondary hospitals (Figure 1C, Supplementary Table A3). The incidence of CDI with presumed origin 'community' increased over the study period, from 0.53 (95% CI: 0.49–0.58) in 2013 to 0.82 (95% CI: 0.75–0.87) per 10,000 patient-days in 2024, while the incidence of cases with presumed origin 'healthcare facility' decreased slightly. (Figure 1B, Supplementary Table A3).



**Figure 2.** Distribution of the 15 most common *Clostridioides difficile* ribotypes analysed by the NRC in Belgium: (A) aggregate data (2013–2024), (B) hospital type (primary [ $N=816$ ], secondary [ $N=177$ ], tertiary [ $N=260$ ]) in 2023–2024, and (C) hospital-associated CDI (HA-CDI,  $N=571$ ) vs non-HA-CDI ( $N=545$ ) in 2023–2024.

CDI: *Clostridioides difficile* infection; HA-CDI: hospital-associated CDI; NRC: National Reference Centre; PCR: polymerase chain reaction.



## CDI microbiology

Over 60 RTs were observed annually, 15 of which accounted for 70% of all strains. Continuous efforts are ongoing to align the nomenclature with the European database. Currently, among the RTs circulating in Belgium, 135 RTs in Belgium have a corresponding European classification, while 27 remain unmatched. Between 2013 and 2024, the four most prevalent RTs nationwide, accounting for approximately one-third of all isolates, were RT014, RT020, RT002, and RT078. RT106 (2.8% in 2013 vs 7.9% in 2024), RT154 (0.3% in 2013 vs 3.7% in 2024), and RT430 (0.1% in 2013 vs 2.5% 2024) emerged significantly in recent years (Figure 2, Supplementary Table A4). The spread of RT027 was controlled, with only one strain reported in 2023 and none in 2024, compared with 55 in 2016.

In the last two years (2023–2024), 13.5% ( $N=188/1392$ ) of isolates harboured binary toxin genes primarily due to RT023, RT078, and RT126, which together accounted for 88.3% of binary toxin–positive strains. All these strains were positive for *tcdA*, *tcdB*, and binary toxin genes (*cdtA/cdtB*). Since 2013, only seven RT033 strains (*tcdA*–, *tcdB*–, and *cdtA/cdtB*+) have been characterised at the NRC, testing negative in cytotoxicity assays. In 2023–2024, most isolates were submitted by primary hospitals ( $N=816$ , 65.1%), while smaller contributions came from tertiary ( $N=260$ , 20.7%) and secondary hospitals ( $N=177$ , 14.1%). RT distribution showed minor variations across hospital types. RT001 and RT154 were more frequent in tertiary hospitals, whereas RT002 and RT078 appeared slightly more common in primary hospitals (Figure 2B). Similarly, only modest differences were observed between HA- and non-HA-CDI cases, with RT005, RT014, and RT106 less frequent in non-HA-CDIs, while RT001 and RT023 showed slight predominance in non-HA-CDIs (Figure 2C).

Between 2013 and 2024, 811 of 3061 (26.5%) recurrent cases, 1323 of 4225 (31.3%) complicated cases, and 185 of 611 (30.3%) CDI-related death cases were linked to a sample sent to the NRC. RT014, RT020, RT002, and RT078 were most frequently identified in recurrent, complicated, and fatal CDI, reflecting their predominance in the overall CDI population (Supplementary Table A5). However, several RTs were significantly over-represented in severe outcomes compared with all other RTs. Among hypervirulent strains, RT027 was the only lineage consistently associated with increased severity, showing significant over-representation in recurrence (12%), complicated CDI (20%), and CDI-related deaths (4%) while being rarely detected in recent years. RT078 and RT023 were significantly more frequent in complicated CDI, with about 20% of isolates per RT linked to complicated cases. Interestingly, among non-hypervirulent RTs, the rare RT087 showed a high proportion of complicated cases (22%), while RT015 and RT002 were over-represented in recurrent CDI. Conversely, RT005, RT012, RT014, and RT017 were significantly under-represented in recurrent and/or complicated CDI, with RT014 reported in fewer CDI-related deaths (Supplementary Table A5).

## Antimicrobial susceptibilities

Since 2019, antimicrobial susceptibility testing (AST) data have been collected for *C. difficile* strains, focussing on antibiotics either associated with CDI development or used in its treatment (Supplementary Table A6). AST results were available for 718 strains, representing 72 RTs collected between

2019 and 2024. All isolates were susceptible to antibiotics currently used in CDI treatment, with mean MICs of 0.5 mg/L for vancomycin, 0.25 mg/L for metronidazole, and 0.0625 mg/L for fidaxomicin. Resistance rates were 93.7% for clindamycin, 16.3% for erythromycin, 11.1% for moxifloxacin, 5.8% for tetracycline, 1.7% for chloramphenicol, and 1.7% for rifampicin. Resistance to multiple antimicrobials (clindamycin, moxifloxacin, tetracycline, and erythromycin) was significantly more frequent in RT012, RT017, RT027, RT126, and RT078 than in other RTs ( $P < 0.05$ ).

## Discussion

The surveillance of CDI in Belgian hospitals is well established, highlighting trends on both epidemiological and microbiological levels with data spanning over more than a decade. The results show several changes over time, including a decrease in the proportion of HA-CDI and complicated infections, an increase in CDI incidence (particularly in recent years), and the eradication of the hypervirulent RT027. Variations in incidence and RT distribution were evident across hospital types. All isolates remained susceptible to the antibiotics currently in use.

Even after it ceased being mandatory in 2015, the surveillance maintained a high participation rate. Unfortunately, hospitals were less compliant with the microbiological component. From our observations, reminding hospitals to send samples did not significantly change their engagement in this respect. In response to a national information request in 2021, laboratories reported that they no longer routinely perform *C. difficile* cultures due to the increasing use of direct stool testing by enzyme immunoassays and nucleic acid amplification tests for the CDI diagnosis, which may help explain the decline in culture submissions to the NRC. Alternative methods need to be considered to compensate for the decrease in the number of samples. Belgium could consider establishing a sentinel network to complement the national surveillance, as has been done in Denmark [16].

In Europe, the CDI epidemiology has evolved in the last decade. Incidences decreased, especially during the COVID-19 pandemic, likely due to improved infection control measures and antimicrobial stewardship [3,17]. In Belgium, while the incidence remained below the European Union/European Economic Area (EU/EEA) average in 2018–2020, the recurrence rate was higher in 2018 and 2019, exceeding 10% compared with 7% reported at the European level. However, the proportion of complicated CDI was below the EU/EEA average for that study period. A considerable increase in missing data was reported in the 2018–2020 ECDC report [18].

The rise in CDI incidence in the last two study years, mainly due to more non-HA-CDI, suggests that more patients are entering the hospital with existing infections, which might strain the healthcare system. It is difficult to confirm whether these infections are due to an increase in CA-CDI or if they are coming from other healthcare facilities. During the surveillance period, an increase was observed in the incidence of CDI with presumed origin 'community'. Given the high percentage of unknown values for this variable, the source of the infection outside the hospital remains uncertain. However, this shift toward CA-CDI was described in North America, challenging the traditional view of CDI as primarily hospital-acquired [19].

Surveillance strategies will need to be adapted to detect these infections. Germany, for example, has emphasised the importance of including CA-CDI in their monitoring systems [20].

Even though incidence rates are generally decreasing in Europe, projections for 2040 indicate an increase in CDI burden primarily due to changes in resistance patterns, the rise of CA-CDI, the emergence of hypervirulent strains, a growing population, and a higher proportion of elderly people [21]. This underlines the importance of maintaining robust surveillance systems.

Between 2013 and 2024, tertiary hospitals consistently reported higher incidence rates than primary and secondary hospitals, a trend also observed in other countries [18,22]. The decline in CDI and HA-CDI incidence in tertiary hospitals in 2020 may have been due to a decrease in care-seeking behaviour during that period. The COVID-19 pandemic has affected the recorded incidence of a wide range of infections, with decreased notifications across almost all notifiable diseases during the pandemic compared with previous years [23]. A higher percentage of severe CDI cases were found during the second wave of COVID-19, likely due to decreased visits to physicians or hospitals, resulting in delayed diagnosis [24].

Molecular surveillance of *C. difficile* over the past decade has revealed substantial shifts in RT distribution and clinical severity. Hypervirulent RT027 was almost eradicated in Belgium, coinciding with the reduction in CDI-related mortality [12]. Although this decline was likely due to multiple factors, including improved diagnostics, infection prevention, and antimicrobial stewardship, the disappearance of RT027 may have played a key role. In contrast, the combatting bacterial resistance in Europe - *Clostridium Difficile* infections study reported that RT027 remained frequently detected, particularly in Eastern Europe [25,26]. The ECDC-coordinated surveillance reported that RT027 accounted for 1.6% of 2838 isolates, alongside RT014/020 (17.7%), RT002 (14.5%), RT078 (7.7%), and RT001 (3.9%) in 2018–2019 [18]. Belgium displays a similar predominance of RT014/020, RT002, and RT078. RT106 has become increasingly prevalent and is one of the predominant RTs in the USA [27], while RT154 remains largely restricted to Belgium and has only been sporadically detected across Europe [18,25,26]. In contrast, several epidemic clones linked to outbreaks elsewhere in Europe, such as RT176, RT181, RT244, RT826, and RT955, do not appear to circulate in Belgium [7].

Additionally, 13.5% of isolates tested positive for the binary toxin in Belgium, which was much lower than the European level (40.9% in 2020) [18]. However, this result is likely biased and underestimated as hospitals participating in the surveillance usually submit only isolates testing positive for TcdA and/or TcdB. These findings highlight the role of local epidemiological dynamics in shaping distinct RT profiles across Europe.

Our findings confirm that hypervirulent RTs such as RT027 and RT078 were disproportionately represented among severe outcomes, consistent with previous European reports [7]. RT023 has also been associated with clinical presentations comparable with those of RT027 and RT078/126 [28]. Interestingly, the over-representation of RT087 in complicated cases reinforced earlier observations that strain severity does not necessarily reflect overall prevalence [29]. Experimental data have shown that RT087 can be as virulent as RT027 in mouse models, despite the absence of classical hypervirulence

markers [30]. Unlike epidemic RTs such as RT023, RT027, and RT078/126, RT087 harbours *tcdA* and *tcdB* but lacks binary toxin genes and *tcdC* deletions and does not appear to be associated with antimicrobial resistance. The determinants underlying its virulence therefore remain unclear. RT087 has also been sporadically reported in Europe, including in Slovenia in 2018 and the Netherlands in 2019 [18]. A recent investigation in the Netherlands reported an increase in RT002 over time, with lower disease severity in recent years [31]. In our setting, no increase in RT002 incidence was observed, and RT002 was not associated with complicated outcomes but was over-represented among recurrent CDI. These findings reinforce the importance of monitoring both hypervirulent lineages and emerging or rare RTs with disproportionate clinical impact.

The mean MICs observed in our study were consistent with those reported in the COMBACTE-CDI study [25]. Although the ESCMID guidelines recommend vancomycin or fidaxomicin as a first-line therapy, metronidazole remains commonly used in Belgium, likely due to reimbursement limitations that restrict access to fidaxomicin [8,32]. This continued reliance on metronidazole may partly explain the slightly higher recurrence rates observed than the European average [18]. While resistance to metronidazole and vancomycin remains low in Europe, reduced susceptibility has been reported in several RTs, including RT027, RT001/072, RT017, RT078, RT106, and RT356/607 [7]. These findings raise concerns about possible treatment failures.

Belgium continues to report lower resistance rates than most European countries, with the notable exception of clindamycin [7]. International studies have confirmed that resistance to clindamycin and fluoroquinolones is widespread in strains such as RT017 and RT027 [33]. In the ECDC 2018–2020 report, 20.3% of *C. difficile* isolates tested across Europe were non-susceptible to moxifloxacin, compared with only 11.8% in our national dataset from 2019 to 2024 [18].

A key strength of this hospital-based surveillance, based on the ECDC protocol, is the consistent and complete case-level data over many years. A limitation is the low number of samples sent, with some hospitals sending more than the requested five. Additionally, overlap between the definitions of complicated CDI and CDI-related death may have led to an over-estimation of the association between certain RTs and severe outcomes.

In conclusion, this study highlighted shifts in CDI characteristics, incidence, RT distribution and diversity, clinical severity, and antimicrobial resistance in Belgium. Given the evolving epidemiology and the emergence of virulent and resistant RTs, sustained efforts to strengthen surveillance are essential. Encouraging hospital participation and ensuring consistent submission of isolates or stool samples will be critical for maintaining surveillance quality. Improved reporting, particularly on infection control measures and treatment regimen, would help clarify patterns in CDI evolution and recurrence. Increasing attention must also be paid to non-HA-CDI, which is becoming a more prominent aspect of the disease burden. Finally, the integration of whole-genome sequencing could provide insights into transmission dynamics, the emergence of hypervirulent or resistant strains, and outbreak investigations [7]. Improved diagnostics and standardised surveillance will enhance cross-country comparability and strengthen disease prevention [17].



## CRedit authorship contribution statement

**M. Callies:** Writing – review & editing, Writing – original draft, Visualisation, Validation, Formal analysis, Data curation. **K. Mertens:** Writing – review & editing, Supervision. **E. Ngyuvula Mantu:** Writing – review & editing, Formal analysis. **K. Soumilion:** Writing – review & editing, Formal analysis. **L. Vaes:** Writing – review & editing, Validation, Project administration, Formal analysis, Conceptualisation. **L. Mortgat:** Writing – review & editing, Validation, Formal analysis, Data curation. **M. Delmée:** Writing – review & editing, Supervision, Conceptualisation. **H. Rodriguez-Villalobos:** Writing – review & editing, Supervision. **B. Catry:** Writing – review & editing, Supervision. **B. Kabamba-Mukadi:** Writing – review & editing, Supervision. **A. Anantharajah:** Writing – review & editing, Writing – original draft, Visualisation, Validation, Formal analysis, Data curation.

## Ethics statement

No ethical approval was needed for this study as it uses routine surveillance data falling under the protection of a Royal Decree. Patient data are pseudonymised through a secure environment following the General Data Protection Regulation.

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## Conflict of interest statement

The authors declare that there are no conflicts of interest (financial, personal, or otherwise).

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## Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jhin.2026.02.011>.

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